



CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023-2024)

NEET(UG)
MINOR
23-04-2023

ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **24** pages.

E2

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

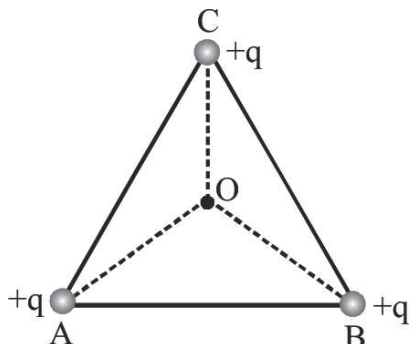
Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

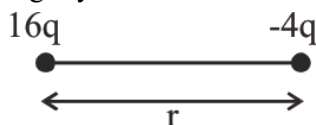
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SECTION-A (PHYSICS)

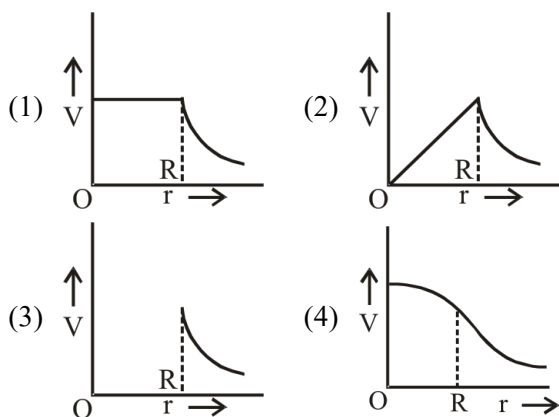
1. Like charges of equal magnitude q are kept at the vertices of an equilateral triangle ABC as shown in the fig. The electric intensity E and potential V , at the centroid O , are respectively :-



- (1) $E = 0, V = 0$
 (2) $E = 0, V \neq 0$
 (3) $E \neq 0, V = 0$
 (4) $E \neq 0, V \neq 0$
2. The electrostatic potential at the neutral point of the point charge system shown will be :-



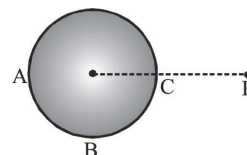
- (1) $\frac{12kq}{r}$ (2) $\frac{42kq}{r}$
 (3) $\frac{4kq}{r}$ (4) $\frac{5kq}{r}$
3. The electric potential due to a uniformly charged non-conducting solid sphere of radius R as a function of the distance from its centre is represented graphically by :-



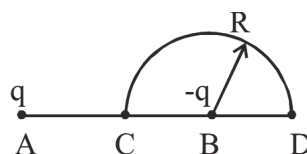
4. What is the electric potential at a distance ' x ' from the centre, inside a conducting sphere of charge ' Q ' and radius ' R ' is :-

- (1) $\frac{KQ}{x}$ (2) $\frac{KQ}{x^2}$
 (3) $\frac{KQ}{R}$ (4) zero

5. A hollow conducting sphere is placed in an electric field produced by a point charge placed at P as shown in figure. Let V_A, V_B, V_C be the potentials at points A, B and C respectively. Then :-



- (1) $V_C > V_B$
 (2) $V_B > V_C$
 (3) $V_A > V_B$
 (4) $V_A = V_C$
6. A conducting sphere of radius R is charged to a potential of V volt. Then the electric field at a distance r ($r > R$) from the centre of the sphere would be :-
- (1) $\frac{V}{r}$ (2) $\frac{R^2V}{r^3}$
 (3) $\frac{RV}{r^2}$ (4) $\frac{rV}{R^2}$
7. Charges $+q$ and $-q$ are placed at points A and B respectively which are a distance $2L$ apart. C is midpoint between A and B . The work done in moving a charge $+Q$ to along the semicircle CRD is :-



- (1) $\frac{qQ}{2\pi\epsilon_0 L}$ (2) $\frac{qQ}{6\pi\epsilon_0 L}$
 (3) $\frac{-qQ}{6\pi\epsilon_0 L}$ (4) $\frac{qQ}{4\pi\epsilon_0 L}$

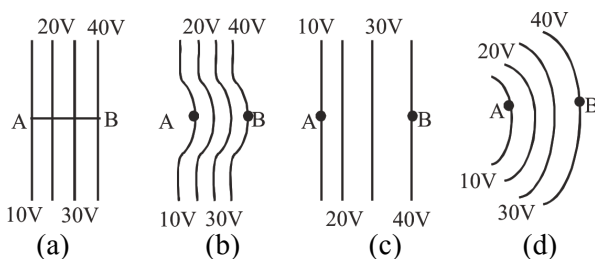
8. List I gives certain situations in which electric field is represented by electric lines of forces in x-y plane. List II gives corresponding representation of equipotential lines in x-y plane. Match the figures in List I with the figures in List II and indicate your answer.

List - I		List - II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

P Q R S

- (1) 1 2 3 4
 (2) 4 3 2 1
 (3) 3 4 2 1
 (4) 2 1 3 4

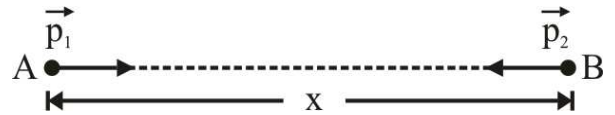
9. The diagrams below show regions of equipotentials:-



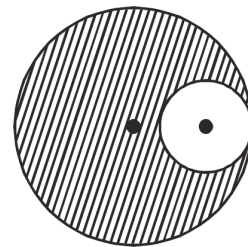
A positive charge is moved from A to B in each diagram.

- (1) In all the four cases the work done is the same
 (2) Minimum work is required to move q in figure (a)
 (3) Maximum work is required to move q in figure (b)
 (4) Maximum work is required to move q in figure (c)

10. Two small electric dipoles, one of dipole moment $\vec{p}_1$ at point A and the other of dipole moment $\vec{p}_2$ at point B, are as shown in the figure. The torque experienced by the dipole $\vec{p}_2$ is :-



- (1) Zero (2) ∞
 (3) Can't determined (4) None of these
11. A cavity is made inside a negatively charged dielectric sphere of volume charge density $+\rho$. Which of the following field pattern is correct in the cavity?

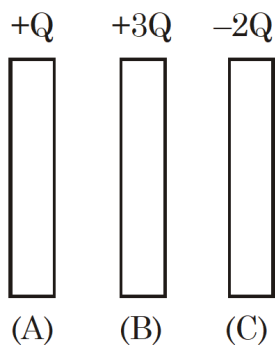


- (1)
- (2)
- (3)
- (4)

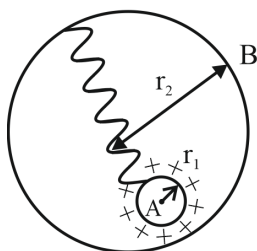
12. A positive point charge 'Q' is brought near an isolated metal cube.

- (1) The cube becomes positively charged.
 (2) The cube becomes negatively charged.
 (3) The interior becomes positively charged and surface becomes negatively charged.
 (4) The interior remains charge free and surface gets non-uniform charge distribution.

13. The ratio of charge on left surface of plate A to the charge on the right surface of plate B is :-



- (1) 0 (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $-\frac{1}{3}$
14. A metal sphere A of radius r_1 charged to a potential ϕ_1 is enveloped by a thin walled conducting spherical shell B of radius r_2 . Then potential ϕ_2 of the sphere A after it is connected to the shell B by a thin conducting wire will be :-



- (1) $\phi_1 \frac{r_1}{r_2}$ (2) $\phi_1 \left(\frac{r_2}{r_1} \right)$
 (3) $\phi_1 \left(1 - \frac{r_2}{r_1} \right)$ (4) $\phi_1 \left(\frac{r_1 r_2}{r_1 + r_1} \right)$
15. A thin conducting plate 50 cm on a side lies in the x-y plane. If a total charge of 4×10^{-8} C is placed on the plate find charge density on the plate.
- (1) 16×10^{-8} C/m² (2) 8×10^{-8} C/m²
 (3) 4×10^{-8} C/m² (4) None
16. A proton is released from rest in a uniform electric field of 8×10^4 V/m $\hat{i}$. The proton undergoes a displacement of 0.5 m in the direction of electric field. Find the change in potential energy of the proton for this displacement.
- (1) $+6.4 \times 10^{-15}$ J (2) -6.4×10^{-15} J
 (3) $+3.2 \times 10^{-15}$ J (4) -3.2×10^{-15} J

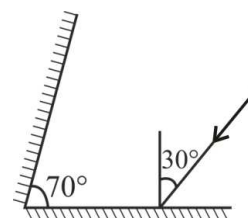
17. Two concentric spherical shells are given charges :-

(A) Outer shell will be at higher potential if the charges given to both shells are negative.
 (B) Inner shell will be at higher potential if charges given to both shells are positive.
 (C) Outer shell will always be at higher potential irrespective of charges given to two sphere.

- (1) B (2) AB (3) C (4) ABC
18. Light rays falls on a plane mirror and forms a real image, so the incident rays are :-

(1) Parallel (2) Diverging
 (3) Converging (4) Statement is false

- 19.



Find angle of deviation after two reflections.

- (1) 120° (2) 220° (3) 240° (4) 115°
20. Two plane mirrors are inclined at 70°. A ray incident on one mirror at angle θ after reflection falls on the second mirror and is reflected from there, parallel to the first mirror, θ is :-
- (1) 50° (2) 45° (3) 30° (4) 55°
21. A plane mirror is approaching you at a speed of 10 cm/s. You can see your image in it. At what speed will your image approach you :-
- (1) 10 cm/s (2) 5 cm/s
 (3) 20 cm/s (4) 15 cm/s
22. A plane mirror is moving with velocity $4\hat{i} + 5\hat{j} + 8\hat{k}$. A point object in front of the mirror moves with a velocity $3\hat{i} + 4\hat{j} + 5\hat{k}$. Here $\hat{k}$ is along the normal to the plane mirror and facing towards the object. The velocity of the image is:
- (1) $-3\hat{i} - 4\hat{j} + 5\hat{k}$ (2) $3\hat{i} + 4\hat{j} + 11\hat{k}$
 (3) $-3\hat{i} - 4\hat{j} + 11\hat{k}$ (4) $7\hat{i} + 9\hat{j} + 11\hat{k}$

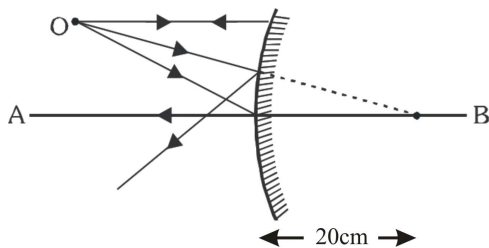
23. An object is placed at 20 cm from a convex mirror of focal length 20 cm. The distance of the image from the pole of the mirror is :

(1) infinity (2) 10 cm
(3) 15 cm (4) 40 cm

24. A concave mirror has focal length 20 cm. The distance between the two positions of the object for which the image size is double of the object size is :-

(1) 20 cm (2) 40 cm (3) 30 cm (4) 60 cm

25. In the figure a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the diagram? (AB is its principal axis) :-



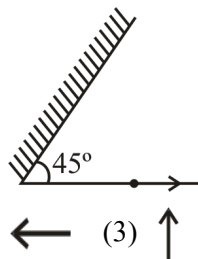
(1) 3 (2) 2 (3) 1 (4) 0

26. **Assertion** : The image formed by a concave mirror is certainly real if the object is virtual.

Reason : The image formed by a concave mirror is virtual if the object is real.

- (1) Both Assertion & Reason are correct & reason is a correct explanation of Assertion.
(2) Both Assertion & reason are correct but reason is not a correct explanation of the assertion.
(3) Assertion correct & Reason false.
(4) Both Assertion & Reason are false.

27. Image of object can be :-

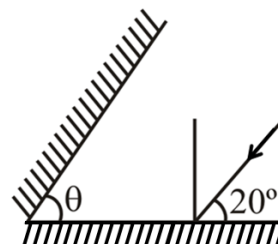


(1) → (2) ← (3) ↑ (4) ↓

28. A plane mirror makes 50° angle with horizontal, a vertical ray strikes the mirror then angle made by reflected ray with mirror will be :-

(1) 50° (2) 40° (3) 90° (4) None

29. In given situation if light ray retraces the path after reflection from second mirror then angle between mirrors will be :-

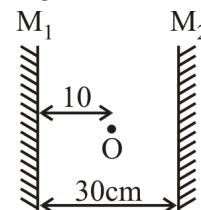


(1) 20° (2) 40° (3) 90° (4) 70°

30. Which of the following alphabates do not show lateral inversion.

(1) A,C,D (2) M,O,N
(3) T,U,V (4) N,O,T

31. Position of 4th image will be :-

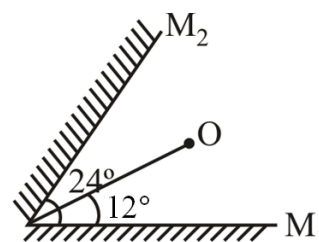


(1) 100 cm from M_2 (2) 110 cm from M_1
(3) 100 cm from M_1 (4) 110 cm from M_2

32. A man of height 1.5 m observes his erect image of 1 m height then mirror used is :-

(1) Concave (2) Convex
(3) Plane (4) Both (1) & (2)

33. In given situation number of images will be :-

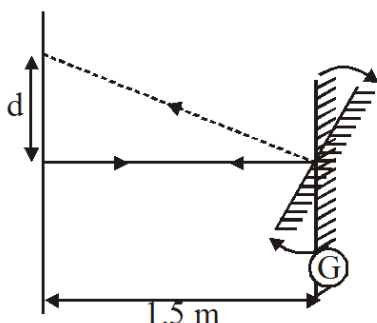


(1) 15 (2) 14
(3) 24 (4) None

34. A divergent light rays are incident on a convex mirror then image formed will be :-

- (1) Real (2) Virtual
(3) Can be both (4) None

35. Light incident normally on a plane mirror attached to a galvanometer coil retraces backwards as shown in figure. A current in the coil produces a deflection of 3.5° of the mirror. What is the displacement of reflected spot of light on a screen placed 1.5 m away ?



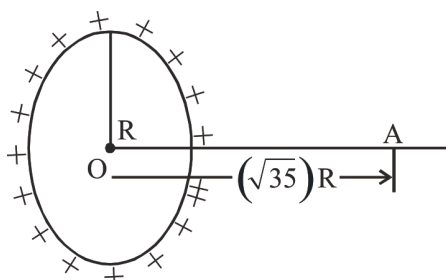
- (1) 10.5 cm (2) 18.31 cm
(3) 15 cm (4) 3 cm

SECTION-B (PHYSICS)

36. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre.

- (1) $\frac{R}{3}$ (2) R (3) $\frac{R}{2}$ (4) None

37.



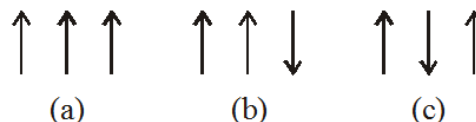
A ring has total charge Q . Find work done by external agent to move $-Q$ charge from point A to centre of ring :-

- (1) $\frac{5KQ^2}{6R}$ (2) $\frac{-5}{6} \frac{kQ^2}{R}$
(3) $\frac{3}{2} \frac{kQ^2}{R}$ (4) $\frac{kQ^2}{4R}$

38. Two concentric spheres of radii R and r have similar charges with equal surface densities (σ). What is the electric potential at their common centre ?

- (1) $\frac{\sigma}{\epsilon_0}$ (2) $\frac{R\sigma}{r\epsilon_0}$
(3) $\frac{\sigma}{\epsilon_0}(R+r)$ (4) $\frac{\sigma}{\epsilon_0}(R-r)$

39. Three identical short electric dipole are placed identical in all situation shown in the diagram.



If U_a , U_b and U_c are interactional potential energy of system a, b and c respectively then:-

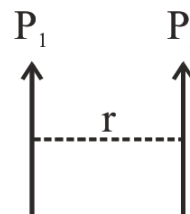
- (1) $U_a > U_b > U_c$
(2) $U_a < U_b < U_c$
(3) $U_a = U_b = U_c$
(4) $U_a < U_c < U_b$

40. **Assertion** :- Under electrostatic conditions net electric field inside a solid conductor will be zero.

Reason :- under electrostatic conditions there will be no free electrons inside a conductor.

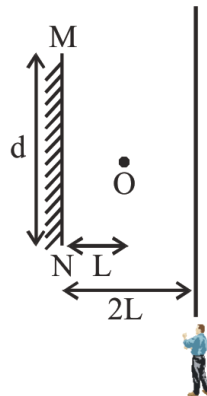
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

41. Two short dipoles are placed parallel to each other as shown in figure. Find force on P_2 due P_1 .

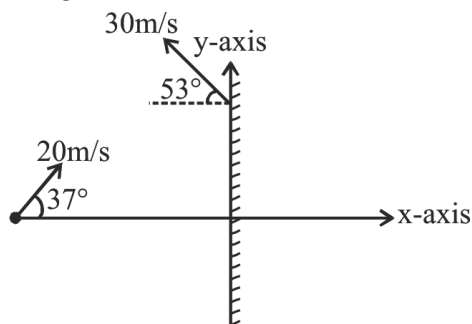


- (1) $\frac{-6KP_1P_2}{r^4}$ (2) $\frac{6KP_1P_2}{r^4}$
(3) $\frac{3KP_1P_2}{r^4}$ (4) $\frac{-3KP_1P_2}{r^4}$

42. A charge $q_1 = 2\mu\text{C}$ is located at the origin, and a charge $q_2 = -6\mu\text{C}$ is located at $(0, 3\text{m})$. Find electric potential due to these charges at the point P, whose co-ordinates are $(4, 0)$ m.
- (1) $6.29 \times 10^3 \text{ V}$ (2) $-6.29 \times 10^3 \text{ V}$
 (3) $3.14 \times 10^3 \text{ V}$ (4) $-3.14 \times 10^3 \text{ V}$
43. Two point charges $2\mu\text{C}$ and $8\mu\text{C}$ are placed at position $x = 0$ and $x = 4\text{m}$ respectively. The distance of the neutral point from the smaller charges is found to be $x = \alpha/\beta$. Find the value of $(\alpha + \beta)$.
- (1) 7 (2) 5 (3) 4 (4) 3
44. A point source of light 'O' is placed at a distance 'L' in front of the centre of a mirror of length 'd' hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it. The distance over which he can see image of light source in the mirror.

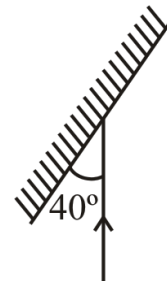


- (1) $d/2$ (2) d (3) $2d$ (4) $3d$
45. Find the velocity of the image with respect to ground of a moving object in situation as shown in figure :



- (1) $12\hat{j}$ (2) $-52\hat{j} - 52\hat{j}$
 (3) $-52\hat{i} + 12\hat{j}$ (4) $20\hat{i} + 12\hat{j}$

46. Under which of the following conditions will a convex mirror of focal length f produce an image that is erect, diminished and virtual for real object :-
- (1) Only when $2f > u > f$
 (2) Only when $u = f$
 (3) Only when $u < f$
 (4) Always
47. The minimum distance between the object and its real image for concave mirror is
- (1) f (2) $2f$
 (3) $4f$ (4) Zero
48. A watch shows time 4 : 20. When seen through mirror image will show time :-
- (1) 8 : 40
 (2) 4 : 20
 (3) 7 : 20
 (4) 7 : 40
49. Find the value of deviation in given diagram.



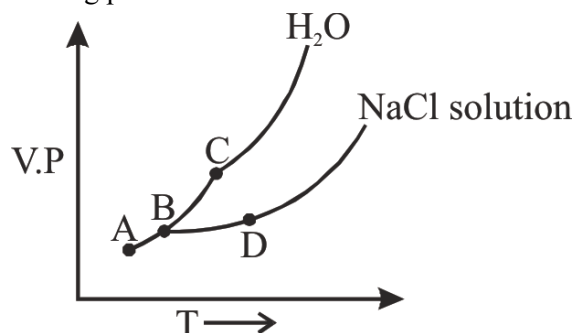
- (1) 100°
 (2) 80°
 (3) 40°
 (4) None
50. A mirror of focal length 30 cm forms image of size '5' times of object then distance of object is :-
- (1) 24 cm
 (2) 36 cm
 (3) Both
 (4) None

Topic : OPTICAL ISOMERISM, CONFORMATIONAL ISOMERISM, INDUCTIVE EFFECT AND MESOMERIC EFFECT, SOLUTIONS (COMPLETE) TG: @Chalnaayaaar

SECTION-A (CHEMISTRY)

51. Which of the following salt has the same value of Vant Hoff factor "i" as that of $K_3[Fe(CN)_6]$:
- (1) Na_2SO_4 (2) $Al(NO_3)_3$
(3) $Al_2(SO_4)_3$ (4) $NaCl$
52. The Van't Hoff factor for 0.1 M $Ba(NO_3)_2$ Solution is 2.74. The degree of dissociation is :-
- (1) 91.3% (2) 87%
(3) 100% (4) 74%
53. A 1% (w/v) KCl solution is ionised to the extent of 80%. The osmotic pressure at $27^\circ C$ of the solution will be :-
- (1) 6.95 atm (2) 5.95 atm
(3) 2.71 atm (4) 3.33 atm
54. 0.2 molal aqueous solution of an electrolyte A_3B is 90% ionised. The boiling point of the solution at 1 atm is.
(K_b for $H_2O = 0.52 K Kg mol^{-1}$ & Boiling point of $H_2O = 373.15 K$) :-
- (1) 273.19 K
(2) 374.92 K
(3) 376.4 K
(4) 373.53 K
55. Mole fraction of solvent in 2 molal aqueous solution will be :-
- (1) 0.017 (2) 0.035
(3) 0.65 (4) 0.965
56. At $80^\circ C$ vapour pressure of pure benzene is 750mm Hg. At this temperature vapour pressure is lowered by 10 mm Hg on dissolving 2g non-volatile solute in 156g benzene. Then molar mass of solute is :-
- (1) 148 (2) 74 (3) 296 (4) 370

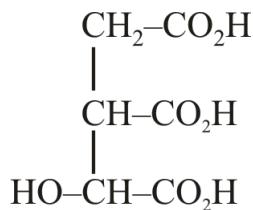
57. A mixture of two volatile substances A and B at a certain temperature shows total vapour pressure $P_T = 100 - 35 X_B$. Values of P_A^0 and P_B^0 respectively are :-
- (1) 100, 65 (2) 100, 35
(3) 65, 100 (4) 100, 135
58. A solution of urea (mol. mass = $60 g mol^{-1}$) boils at $100.18^\circ C$ at the atmospheric pressure. If K_f and K_b for water are 1.86 and $0.512 K kg mol^{-1}$ respectively, the above solution will freeze at :
- (1) $-6.54^\circ C$ (2) $-5.12^\circ C$
(3) $0.654^\circ C$ (4) $-0.654^\circ C$
59. Freezing point of solution is marked as :-



- (1) A (2) B (3) C (4) D
60. Incorrect statement about azeotrope is.
- (1) These are binary mixture having same composition in liquid & vapour phase.
(2) Azeotropic mixtures boils at a constant temperature.
(3) The components of azeotropes can be separated by fractional distillation.
(4) 68% Nitric acid & 32% water by mass form maximum boiling azeotrope.
61. Assuming complete ionisation, the solution having maximum freezing point will be –
- (1) 1 M CaF_2 (2) 1.5 M $Al_2(SO_4)_3$
(3) 2 M $NaCl$ (4) 1 M $AgNO_3$

62. If O_2 gas is bubbled through water at 303 K. The number of millimoles of O_2 gas that dissolve in 1L water at 0.92 bar pressure is (Given : K_H of O_2 at 393 K is 46.82 Kbar.)
 (1) 5 (2) 1 (3) 15 (4) 10
63. Which are correct statements –
 (a) Henry law tells effect of pressure on solubility of gas at given temperature.
 (b) Solubility of gas increases on increasing pressure.
 (c) Solubility of gas increases with increasing temperature.
 (d) Henry's law is not applicable for only non reactive gas.
 (1) a, b (2) b, c (3) c, d (4) a, c
64. Which are colligative properties :
 (1) Boiling point & osmotic pressure
 (2) Freezing point & osmosis
 (3) Elevation in boiling point & osmotic pressure
 (4) Vapour pressure & depression in freezing point
65. During depression in freezing point of a solution the following are in equilibrium.
 (1) liquid solvent $\rightleftharpoons$ Solid solvent
 (2) liquid solvent $\rightleftharpoons$ Solid solute
 (3) liquid solute $\rightleftharpoons$ Solid solute
 (4) liquid solute $\rightleftharpoons$ Solid solvent
66. The vapour pressure of a given liquid will decrease if –
 (1) The liquid is moved to a container in which surface area is very much small.
 (2) The volume of the vapour phase is decreased.
 (3) The volume of the vapour phase is increased.
 (4) Temperature is decreased.
67. A dilute solution of glucose shows depression in freezing point by 1.86 K. K_f for water is 1.86 K kg mol⁻¹. How many moles of glucose are present in 1L of water solvent.
 (1) 0.018 mol (2) 1 mol
 (3) 0.36 mol (4) 2 mol
68. Identify conformer of 2-methyl pentane :-
- (1)
- (2)
- (3)
- (4)
69. Which of the following is most stable conformer of n-butane :-
- (1)
- (2)
- (3)
- (4)

70. Number of chiral center and stereoisomers are :

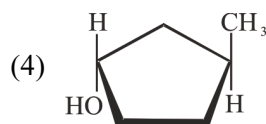
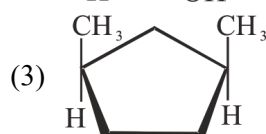
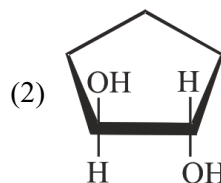
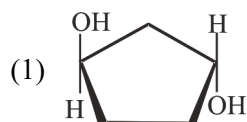


- (1) 1,2 (2) 2,4 (3) 2,3 (4) 3,8

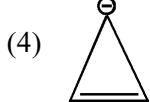
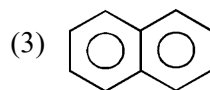
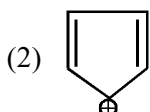
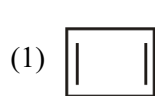
71. Optical isomerism show by :-

- (1) Pentan-1-ol (2) Pentan-2-ol
(3) Pentan-3-ol (4) All of these

72. Which of the following cyclopentane derivative is optically inactive ?



73. Among the following the aromatic compound is –



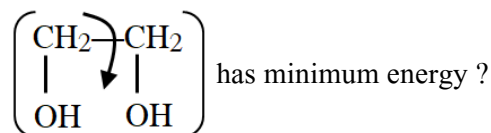
74. Which of the following series contains electrophiles only ?

- (1) H_2O , SO_2 , H_3O^+ (2) NH_3 , H_2O , AlCl_3
(3) AlCl_3 , SO_3 , NO_2^+ (4) H_2O , Cl^+ , NH_3

75. Which of the following can show conformations ?

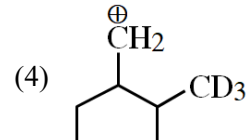
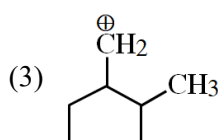
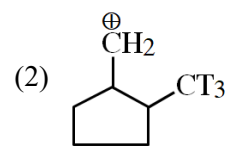
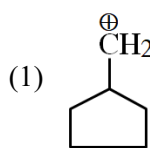
- (1) CH_4 (2) C_2H_4
(3) C_2H_2 (4) C_3H_8

76. Which conformations of ethylene glycol

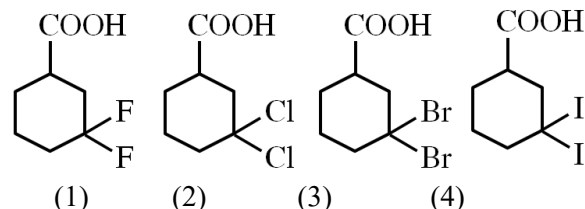


- (1) Gauche
(2) Anti (staggered)
(3) Fully eclipsed
(4) Partial eclipsed

77. Which of the following carbocation is most stable ?

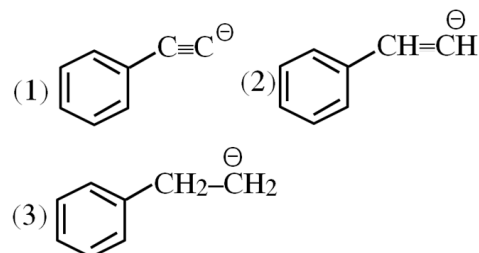


78. Which of the following represents the correct order of acidity in the given compounds ?



- (1) $2 > 1 > 3 > 4$
(2) $4 > 3 > 2 > 1$
(3) $1 > 2 > 3 > 4$
(4) $3 > 2 > 1 > 4$

79. Basic strength order will be

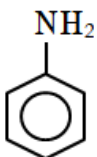
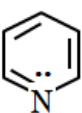
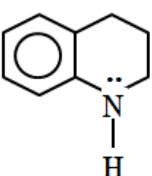
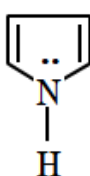


- (1) $1 > 2 > 3$ (2) $2 > 1 > 3$
(3) $3 > 1 > 2$ (4) $3 > 2 > 1$

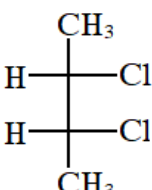
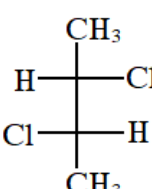
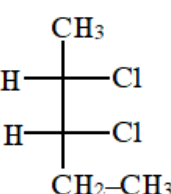
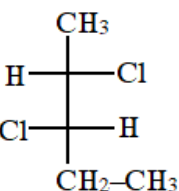
80. Which of the following has lowest pK_a value ?

- (1) $\text{Cl}-\text{CH}_2-\text{COOH}$ (2) $\begin{array}{c} \text{Cl}-\text{CH}-\text{COOH} \\ | \\ \text{Cl} \end{array}$
 (3) $\text{Cl}_3\text{C}-\text{COOH}$ (4) $\text{CH}_3-\text{CO}_2\text{H}$

81. In which of the following compound lone pair of nitrogen is localised ?

- (1)  (2) 
 (3)  (4) 

82. Which of the following is a meso compound ?

- (1) 
 (2) 
 (3) 
 (4) 

83. $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}-\text{CH}_3 \\ | \\ \text{Cl} \end{array}$

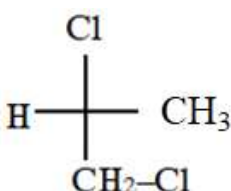
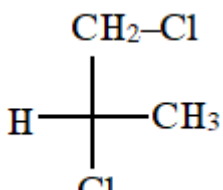
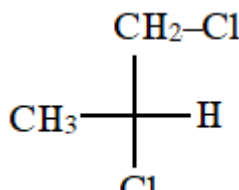
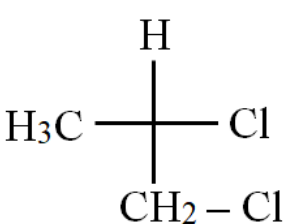
Total number of stereoisomers are :

- (1) 4 (2) 6 (3) 8 (4) 12

84. Which of the following is the correct order of -I effect groups.

- (1) $-\text{NO}_2 > -\text{CN} > -\text{Br}$
 (2) $-\text{Cl} > -\text{NO}_2 > -\text{CN}$
 (3) $-\text{CH}=\text{CH}_2 > -\text{NO}_2 > -\text{OH}$
 (4) $-\text{NH}_2 > -\text{OH} > \text{NO}_2$

85. Which of the following compound has S configuration ?

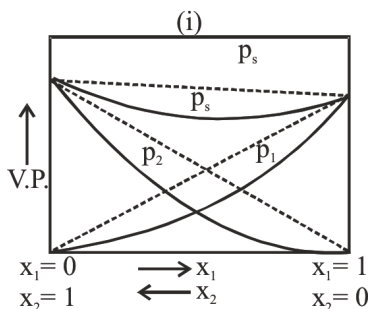
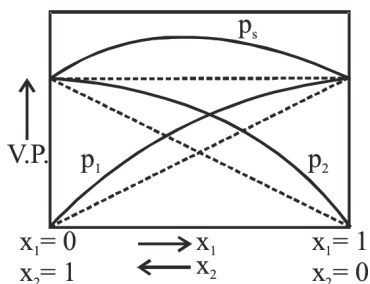
- (1) 
 (2) 
 (3) 
 (4) 

SECTION-B (CHEMISTRY)

86. Which of the following is not correct for an ideal solution ?

- (1) $\Delta G_{\text{mix}} > 0, \Delta S_{\text{mix}} > 0$
 (2) $\Delta H_{\text{mix}} = 0, \Delta V_{\text{mix}} = 0$
 (3) $\Delta V_{\text{mix}} = 0, \Delta S_{\text{mix}} > 0$
 (4) $\Delta H_{\text{mix}} = 0, \Delta G_{\text{mix}} < 0$

87. Study the figure given below and mark the correct statement :-

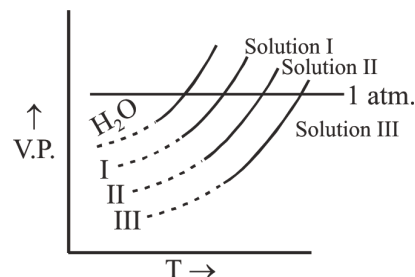


- (1) (i) Nitric acid + Water
(ii) Acetone + Ethyl alcohol
- (2) (i) Water + Ethyl alcohol,
(ii) Acetone + Benzene
- (3) (i) Acetone + Ethyl alcohol
(ii) Acetone + Chloroform
- (4) (i) Benzene + Chloroform
(ii) Acetone + CCl_4
88. Which of the following pair of solutions are expected to be isotonic at the same temperature ?
- (1) 0.2 M urea & 0.2 M NaCl
- (2) 0.1 M urea & 0.2 M MgCl_2
- (3) 0.1 M NaCl & 0.1 M Na_2SO_4
- (4) 0.1 M $\text{Ca}(\text{NO}_3)_2$ & 0.1 M Na_2SO_4
89. Henry's law constant for gases are given below at 293 K. Which one is more soluble in water at this temperature and same pressure :-
- (1) A ($K_H = 34.86$ Kbar)
- (2) B ($K_H = 69.16$ Kbar)
- (3) C ($K_H = 144.97$ Kbar)
- (4) D ($K_H = 88.84$ Kbar)

90. The vapour pressure of benzene and toluene are 42 mm Hg and 36 mm Hg respectively. A solution contains benzene and toluene in the molar ratio of 1 : 2. The mole fraction of benzene in vapour phase is :-

- (1) $\frac{12}{19}$ (2) $\frac{7}{19}$
- (3) $\frac{11}{18}$ (4) $\frac{3}{4}$

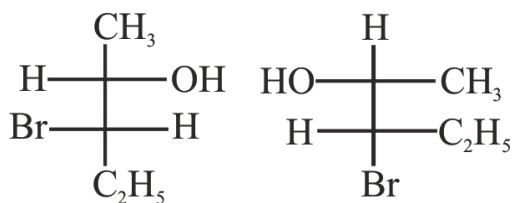
- 91.



Which solution have highest elevation in boiling point :

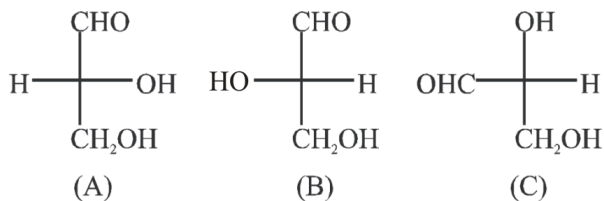
- (1) H_2O
- (2) Solution I
- (3) Solution II
- (4) Solution III
92. **Assertion:** When NaCl is added to water a depression in freezing point is observed.
Reason: The lowering of vapour pressure of a solution causes depression in the freezing point.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.
93. When sugar is added to water its –
- (1) Vapour pressure decreases
- (2) Boiling point increases
- (3) Freezing point increases
- (4) both (1) & (2)

94. What is relationship between these compounds ?



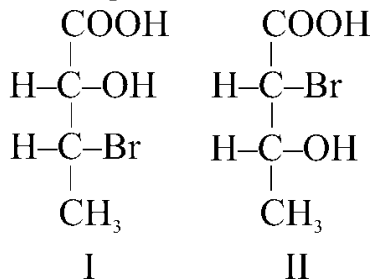
- (1) Identical (2) Diastereomers
(3) Enantiomers (4) Epimers

95. For the given compounds A,B,C identify the statement that is true :-



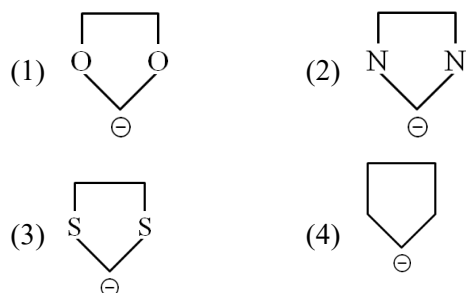
- (1) A and B are identical
(2) A and C are enantiomers
(3) B and C are identical
(4) A and B are enantiomers

96. The compound I & II are related with each other as:-

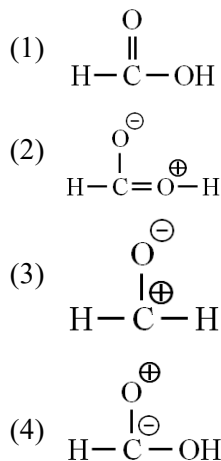


- (1) Position isomers
(2) Enantiomers
(3) Conformational isomers
(4) diastereomers

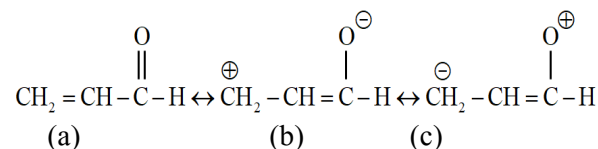
97. Which of the following carbanion is stablized by resonance ?



98. Most stable resonating structure of formic acid ?

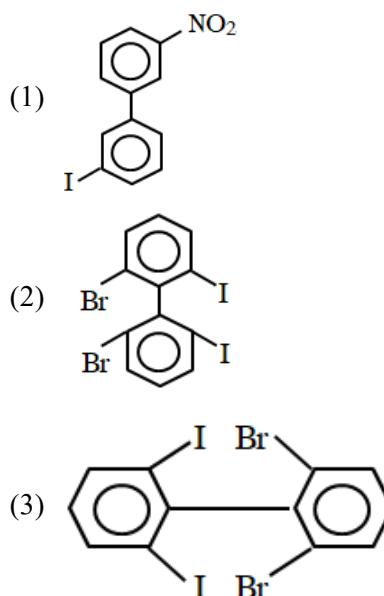


99. Correct order of stability of the following resonating structure ?



- (1) $a > b > c$
(2) $a > c > b$
(3) $b > a > c$
(4) $c > b > a$

100. Which of the following biphenyl is optically active ?



- (4) None of these

Topic : HUMAN HEALTH AND DISEASE (TILL DATE), ECOSYSTEM, BIOMES, BIODIVERSITY AND CONSERVATION

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

101. Good humor hypothesis is disproved by :

- (1) Von behring (2) Louis pasture
(3) William Harvey (4) Aristotle

102. Mucus in urogenital tract includes in

- (1) Physiological barrier (2) Cytokine barrier
(3) Physical barrier (4) Cellular barrier

103. Antibodies are formed by

- (1) Glycolipid (2) Cholesterol
(3) Polysaccharide (4) Glycoprotein

104. Asthma is a :

- (1) Allergic disorder
(2) Auto immune disorder
(3) Hormonal disorder
(4) Vitamin deficiency disorder

105. NK cells provide

- (1) Acquired immunity (2) Active immunity
(3) Innate immunity (4) Passive immunity

106. Concept of vaccine depends upon

- (1) Innate immunity (2) Memory
(3) Physical barrier (4) Cellular immunity

107. Which of the following statement is not true regarding interferons ?

- (1) It is a type of cytokine barrier
(2) Secreted by virus-infected cells
(3) It is made up of fat
(4) It is species specific

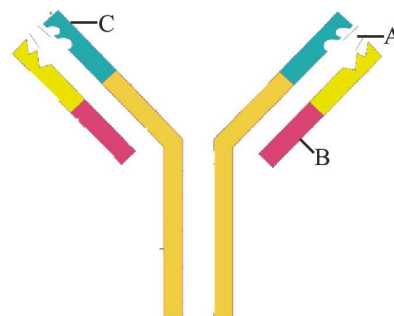
108. Antiserum contain :-

- (1) Antigens
(2) Antigen and Antibody
(3) Antibody against serum
(4) Antibody containing serum

109. Histamine is secreted by :-

- (1) Basophil (2) Mast cells
(3) Neutrophil (4) Both 1 & 2

110. Identify the correct option for given diagram for A, B and C :-



	A	B	C
(1)	Epitope	Light chain	Heavy chain
(2)	Paratope	Constant light chain	Variable heavy chain
(3)	Epitope	Constant light chain	Constant heavy chain
(4)	Paratope	Constant light chain	Constant heavy chain

111. Which of the following statement is **false** ?

- (1) N-K cells are granular cells
(2) inflammation is a local response of living mammalian tissue to injury
(3) AMI takes part in only innate immunity.
(4) N-K cells kill the target cell by creating perforin lined pores in their cell membrane

112. Antibodies are secreted by :-

- (1) T-lymphocytes (2) B-lymphocytes
(3) both (1) and (2) (4) natural killer cell

113. Non-phagocytic cell is :-

- (1) Neutrophil (2) Monocyte
(3) Macrophage (4) N.K. Cell

114. Which antibody is first generated in response to antigen ?
 (1) IgG (2) IgM (3) IgA (4) IgE
115. All 'lymphocytes' are produced in :-
 (1) Thymus (2) Bone marrow
 (3) Spleen (4) Both '1' and '2'
116. Antibodies form against inhaled or ingested pathogen is :
 (1) IgG (2) IgD
 (3) IgA (4) IgM
117. Which of the following cells do not secrete antibodies but stimulate B-lymphocytes for antibody formation :-
 (1) T-cells (2) B plasma cell
 (3) N.K. cell (4) All of these
118. According of good humor hypothesis which person will belong to hot personality ?
 (1) Person with more black bile
 (2) Person with more yellow bile
 (3) Person who has more blood
 (4) Person with good health
119. Health according to W.H.O. is :-
 (1) Physical fitness only
 (2) Physical + Mental fitness only
 (3) Physical + Mental + Social well being
 (4) All of these
120. Which protect non infected cells from further viral infection :-
 (1) Cellular barrier (2) Antibodies
 (3) Cytokine barrier (4) Physical barrier
121. After maturation the lymphocytes migrate to :-
 (1) Bone marrow (2) Thymus
 (3) 1 & 2 both (4) Spleen

122. Memory based acquired immunity involved in _____ based on the differentiate foreign organisms:-

- (1) Lower plants (2) Bacteria
 (3) Higher vertebrates (4) Lower vertebrates

123. Non specific immunity includes all of these except :-

- (1) Immunoglobulin (2) Lysozyme
 (3) Interferons (4) Neutrophils

124. Match the columns with regard to innate immunity :-

Column I		Column II	
(A)	Interferon	(i)	Physiological barrier
(B)	Gastric juice, saliva, tear	(ii)	Physical barrier
(C)	Mucosa and Skin	(iii)	Cellular Barrier
(D)	Neutrophils, monocyte, N-K cells	(iv)	Cytokine Barrier

- (1) A-i, B-ii, C-iii, D-iv
 (2) A-iv, B-i, C-ii, D-iii
 (3) A-iv, B-i, C-iii, D-ii
 (4) A-iii, B-iv, C-i, D-ii

125. Which of the following is true about passive immunity?

- (1) Immune response-Long lasting and more effective
 (2) Immunity develops after lag period
 (3) Immunological memory present
 (4) Immunity responds immediately

126. All are true for colostrum except :-

- (1) It has IgA type antibodies
 (2) It is an example of passive immunity
 (3) Milky whitish fluid secreted by mother during the final days of lactation
 (4) Essential for new born infant

127. Antigen binding sites on an antibody are formed by :

- (1) Variable region of light chain and constant region of heavy chain
- (2) Constant region of light chain and variable region of heavy chain
- (3) Variable region of light chain and variable region of heavy chain
- (4) Constant region of light chain and constant region of heavy chain

128. How many following structures are responsible for formation of second line of defence ?

Skin, Mucosa, Neutrophils, monocytes, macrophages, T-lymphocytes, interferon, fever, B-lymphocytes.

- (1) Three (2) four (3) five (4) six

129. Yoga has been practised from time immemorial to achieve :-

- (1) Physical health (2) Mental health
(3) Both (4) None

130. Incorrect about spleen is :-

- (1) Secondary lymphoid organ
- (2) Contains phagocytes only
- (3) Traps blood-borne micro organisms
- (4) Bean shaped organ

131. Which one of the following immune system components does not correctly match with its role?

(A)	Interferon	Secreted by virus infected cells and protect virus infected cells from further viral infection.
(B)	Macrophage	Large phagocytic cell that phagocytose microbes entering in the body.
(C)	Plasma Cell	Produces antibodies in response to pathogen into blood to fight with them.
(D)	IgA	Present in colostrum in early days of lactation to protect foetus from disease.

- (1) A & C (2) A & D
(3) 'A' only (4) 'D' only

132. Which of the following is not bacterial disease –

- (1) Dysentery (2) Plague
(3) Diphtheria (4) Rhinitis

133. Health is affected by –

- (1) Genetic disorders (2) Infections
(3) Life style (4) All of these

134. Sporozoites are stored in –

- (1) Salivary gland of Human
(2) Salivary gland of female Anopheles
(3) Blood of female Anopheles
(4) RBC of Human

135. Ascariasis spread through –

- (1) Contaminated water
(2) Contaminated fruits
(3) Contaminated vegetables
(4) All of these

SECTION-B (BIOLOGY-I)

136. Which is correct for Health :

- (1) Increase longevity of people
(2) Reduce infant mortality
(3) Reduce maternal mortality
(4) All

137. Which is incorrect for anamnestic response :

- (1) Highly intensified
(2) Secondary response
(3) Concentration of Antibodies more than PIR
(4) Slow response

138. Which is correct for Acquired immunity :

- (1) Character of memory present
(2) Discrimination property present
(3) Diversity present
(4) All of these

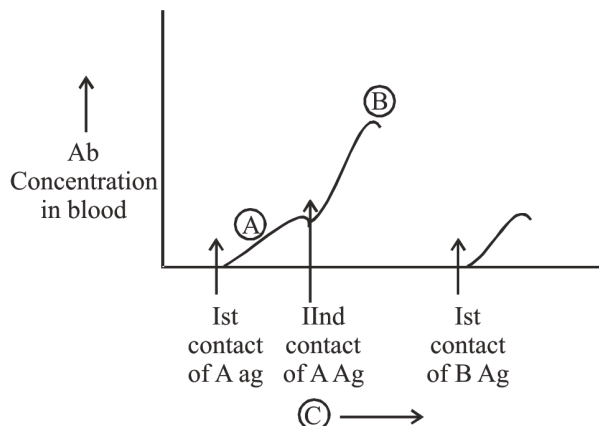
139. Which of the following is not include in physical barrier :-

- (1) Skin (2) Mucus
(3) PMNL (4) Cilia

140. Which physiological process of body create adverse environment for growth of Microbes ?

- (1) Acid in stomach (2) Tears from eyes
(3) Saliva in mouth (4) All of above

141. Identify the correct A,B,C in given diagramme



- (1) A-Primary IR, B-Secondary IR, C-Ab in Blood
(2) A-Secondary IR, B-Primary IR, C-Time period
(3) A-Primary IR, B-Secondary IR, C-Time period
(4) A-Secondary IR, B-Primary IR, C-Ab in Blood

142. Which of the following is not included in cellular barrier of innate immunity ?

- (1) Monocyte
(2) P.M.N.L. neutrophils
(3) T killer cell
(4) Macrophage

143. Which type of bond is present in between light and heavy chain of an antibody?

- (1) Ionic bond (2) Peptide bond
(3) Dioxide bond (4) Disulphide bond

144. Primary response which is of ___(A)___ intensity. Subsequent encounter with the same pathogen elicit a ___(B)___ intensified ___(C)___ response :-

	A	B	C
(1)	High	Low	Anamnestic
(2)	Low	High	Primary
(3)	Low	High	Secondary
(4)	High	Low	Primary

145. Find out the correct statement.

- (1) T-cells secrete antibodies and also help B-cells to produce them.
(2) Innate immunity is pathogen specific.
(3) Fever is an immune response
(4) Antibody provides cellular immunity.

146. Which antibody shows opsonization ?

- (a) IgD (b) IgA (c) IgG
(d) IgM (e) IgE
(1) (a), (b), (c) (2) (c), (d), (e)
(3) (c), (d) (4) (a), (e)

147. Which of the following options gives the correct categorisation of barriers of Innate immunity ?

	Physical	Physiological	Cellular
(1)	N.K. cell	Saliva	PMNL neutrophils
(2)	Skin	Tear	W.B.C.
(3)	Mucous	Interferon	T-killer cells
(4)	Fever	Phagocyte cells	Leukocyte

148. Antimicrobial substances present in –

- (1) Tears (2) Saliva
(3) Both 1 & 2 (4) Mucus

149. Non infectious disease which is major cause of death is –

- (1) Cancer (2) AIDS
(3) Malaria (4) Dengue

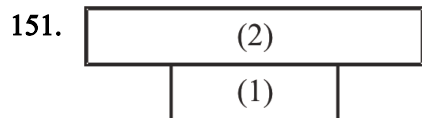
150. Select incorrect option of Entamoeba –

- (1) Causes amoebic Dysentery
(2) Protozoan parasite
(3) Mosquito act as Mechanical carrier
(4) Parasite of Large intestine

Topic : HUMAN HEALTH AND DISEASE (TILL DATE), ECOSYSTEM, BIOMES, BIODIVERSITY AND CONSERVATION

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SECTION-A (BIOLOGY-II)



Above pyramid represent.

- (1) Pyramid of grassland on the basis of biomass.
 - (2) Pyramid of grassland on the basis of number.
 - (3) Pyramid of aquatic ecosystem on basis of biomass
 - (4) Pyramid of energy
152. The annual net primary productivity of the whole biosphere is :-
- (1) 190 billion tons
 - (2) 160 billion tons
 - (3) 170 billion tons
 - (4) 150 billion tons
153. Mammals from colder climates generally have shorter ears and limbs to minimise heat loss, this is
- (1) Bergman's law
 - (2) Allen's law
 - (3) Rensch rule
 - (4) Jordan's rule
154. Which of the following may result in build up of organic materials ?
- (1) Low temperature and anaerobiosis
 - (2) High temperature and aerobiosis
 - (3) Hot and humid conditions
 - (4) Fast humification and fast mineralisation

155. Which of the following is a sedimentary cycle ?

- (1) Carbon cycle
- (2) Nitrogen cycle
- (3) Phosphorous cycle
- (4) Oxygen cycle

156. Amazon Rain forest produce total _____ percent oxygen of the earth's atmosphere.

- (1) 30%
- (2) 40%
- (3) 20%
- (4) 25%

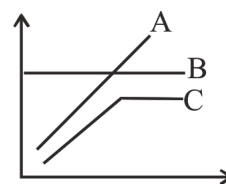
157. Which of the following are the most important climatic factor regulating rate of decomposition:-

- (1) Temperature
- (2) Moisture
- (3) Nature of Detritus
- (4) Both 1 and 2 above

158. The biomes with fragile ecosystem with perma frost condition, having very low rainfall is?

- (1) Tropical rain forest
- (2) Temperate deciduous forest
- (3) Arctic tundra
- (4) Tropical deciduous

159. Identify A, B and C in the graph given below :



- (1) A : Regulator ; B : Conformer ; C : Partial regulator
- (2) A : Regulator ; B : Conformer ; C : Partial conformer
- (3) A : Regulator ; B : Partial conformer ; C : Conformer
- (4) A : Conformer ; B : Regulator ; C : Partial regulator

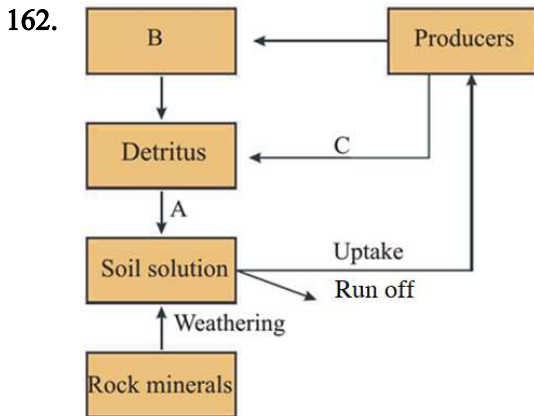
160. Statement -1 :- The function of the reservoir is to meet the deficit which occurs due to imbalance in the rate of influx and efflux

Statement -2 :- Atmospheric inputs of phosphorus through rainfall are almost same as carbon.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

161. Which one of the following is not a functional unit of an ecosystem ?

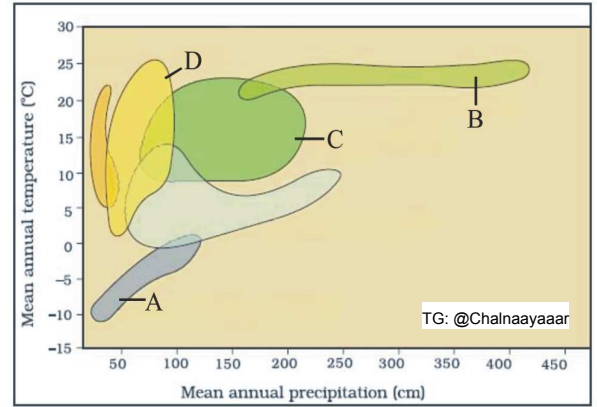
- (1) Energy flow
- (2) Decomposition
- (3) Productivity
- (4) Stratification



In the above phosphorus cycle Identify A, B & C.

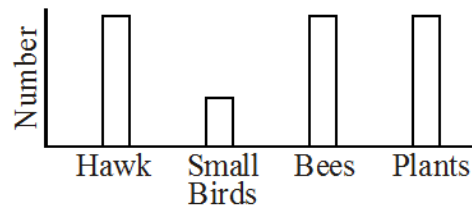
	A	B	C
(1)	Decomposition	Consumer	Litter fall
(2)	Weathering	Producer	Rock
(3)	Decomposition	Consumer	Rock
(4)	Weathering	Decomposer	Litter fall

163. A climograph of various biome is given below. Identify the biome A, B, C, D represents ?

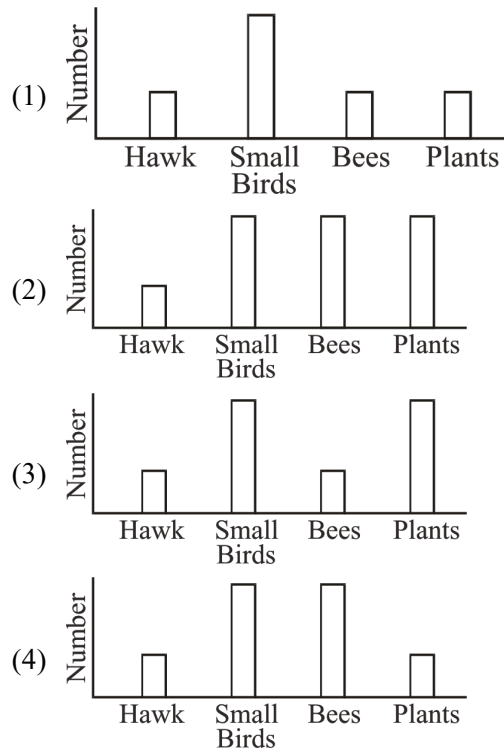


	A	B	C	D
(1)	Tundra	Tropical forest	Temperate forest	Grassland
(2)	Tropical forest	Tundra	Temperate forest	Grassland
(3)	Grassland	Temperate forest	Tundra	Tropical forest
(4)	Grassland	Tropical forest	Temperate forest	Tundra

164. In an ecosystem, four major interdependent components were in the given proportions :



What would be the new composition of the same ecosystem when majority of the Hawk were killed



- 165.** Which of the following is the most important cause for animals and plants being driven to extinction?
- (1) Over exploitation
 - (2) Economic exploitation
 - (3) Alien species invasion
 - (4) Habitat loss and fragmentation
- 166.** Which scientist proposed "Rivet Popper Hypothesis" related to biodiversity and ecosystem?
- (1) Alexander Von Humboldt
 - (2) David Tilman
 - (3) Paul Ehrlich
 - (4) Robert May
- 167.** Joint Forest Management concept was introduced in India in :-
- (1) 1960
 - (2) 1970
 - (3) 1980
 - (4) 1990
- 168.** In India, how many biodiversity Hotspots have been identified Till date ?
- (1) 2
 - (2) 3
 - (3) 5
 - (4) 7
- 169. Statement-I :** Eastern ghats have a greater amphibian species diversity than the western ghats.
Statement-II : Species diversity increase as we go/move away from the equator toward the poles.
- (1) Both statement-I and statement-II are correct.
 - (2) Statement-I is correct but statement-II is incorrect
 - (3) Statement-I is incorrect but statement-II is correct
 - (4) Both statement-I and statement-II are incorrect
- 170.** The region of biosphere reserve where no human activities are allowed is known as :-
- (1) Buffer zone
 - (2) Core zone
 - (3) Transition zone
 - (4) Restoration zone
- 171.** Most animals are tree dwellers in a :-
- (1) Thron forest
 - (2) Temperate deciduous forest
 - (3) Tropical rain forest
 - (4) Coniferous forest
- 172.** The organization which publishes the Red Data Book :-
- (1) IUCN
 - (2) ICFRE
 - (3) WWF
 - (4) FRI
- 173.** The symbol of WWF is :-
- (1) Dolphin
 - (2) Tiger
 - (3) Giant Panda
 - (4) Hornbill
- 174.** An example of ex-situ conservation is :-
- (1) National Park
 - (2) Wildlife Safari Park
 - (3) Wildlife Sanctuary
 - (4) Biosphere Reserve
- 175.** A species facing extremely high risk of extinction is called :-
- (1) Vulnerable
 - (2) Endemic
 - (3) Extinct
 - (4) Critically Endangered
- 176.** Species-Area relationship was described by ?
- (1) Robert May
 - (2) David Tilman
 - (3) Alexander Von Humboldt
 - (4) E.P. Odum

177. Which one of the following is an example of renewable resources?

- (1) Forest (2) Fossil Fuel
(3) Nuclear energy (4) Natural gas

178. Amrita Devi Bishnoi Award instituted by Government of India is given for :-

- (1) Soil conservation
(2) Protecting wildlife
(3) Water conservation
(4) Protecting tribal people.

179. Which one of the following is correctly matched?

- (1) Rhinoceros – Desert National Park
(2) Lion – Manas Wild life Sanctuary
(3) Tiger – Gir National Park
(4) Hangul – Dachigam Sanctuary

180. In India Wild Life (Protection) Act was passed in :-

- (1) 1980 (2) 1972 (3) 1992 (4) 1929

181. Jim Corbett National Park is famous for and is located in ?

- (1) Lion – Uttarakhand
(2) Tiger – Gujarat
(3) Tiger – Uttarakhand
(4) Rhino – Assam

182. The Runn of Kutch in Gujarat famous for :-

- (1) Red Panda (2) Hangul
(3) Asiatic Wild Ass (4) Lion

183. India has more than __A__ genetically different strains of rice and __B__ varieties of mango :

- (1) A-50,000 B-1,000
(2) A-1000, B-50,000
(3) A-5000, B-10,000
(4) A-10,000, B-5,000

184. Which one of the following is representing highest global biodiversity among vertebrates ?

- (1) Mammals (2) Birds
(3) Reptiles (4) Amphibians

185. Exploration of molecular, genetic and species level diversity for products of economic importance is known as :-

- (1) Bio-energetics (2) Bio-prospecting
(3) Bio-remediation (4) Bio-informatics

SECTION-B (BIOLOGY-II)

186. Earth is an :

- (1) Open system for energy and open system for minerals
(2) Closed system for minerals and open system for energy
(3) Closed system for energy and open system for minerals
(4) Closed system for energy and closed system for minerals

187. The amount of Inorganic substances present at a particular area at any given time in an ecosystem is known as :-

- (a) Standing state
(b) Standing crop
(c) Biomass
(d) Primary productivity

- (1) a (2) b (3) b, c (4) b,d

188. "To meet with the deficit which occurs due to imbalance in the rate of influx and efflux" is the function of :-

- (1) Biosphere
(2) Reservoir
(3) Solar energy
(4) Species diversity

189. Choose the correct statement :-

- (a) Atmospheric input of phosphorus through rainfall is much smaller than carbon.
- (b) Plants absorb phosphorus in form of orthophosphate
- (c) Gaseous exchange of phosphorus between organism and environment is negligible
- (d) Atmosphere only contains about 71 percent of total global carbon.

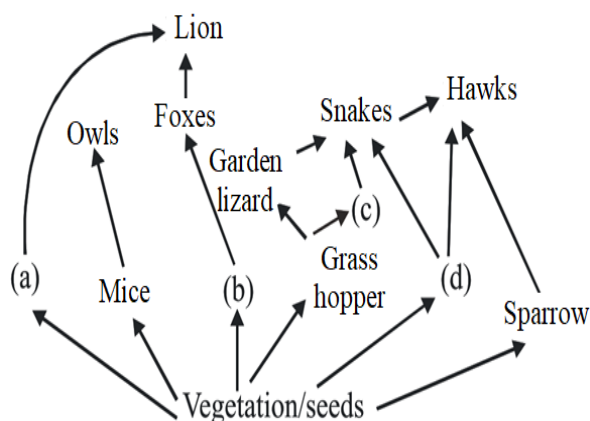
Options :-

- (1) a, b and c are correct
- (2) only a and b are correct
- (3) only b and d are correct
- (4) only a and c are correct

190. Mr. Rahul is eating a cup of curd. What is his trophic position ?

- (1) T₂ (2) T₁ (3) T₅ (4) T₃

191. Identify the likely organisms (a), (b), (c) and (d) in the food web shown below :



Options :

	(a)	(b)	(c)	(d)
(1)	Rat	Dog	Tortoise	Crow
(2)	Squirrel	Cat	Rat	Pigeon
(3)	Deer	Rabbit	Frog	Rat
(4)	Dog	Squirrel	Bat	Deer

192. Consider the following statements about limitations of ecological pyramids :-

- I. It never takes into account the same species belonging to two or more trophic levels.
- II. It assumes a simple food chain, which never exists in nature.
- III. In spite of the vital role played by saprophytes/decomposers, they are not given any position in ecological pyramids.

Which of the statements given above are correct ?

- (1) I and II (2) I and III
- (3) II and III (4) I, II and III

193. **Assertion:** Temperature is the most important ecologically relevant environmental factor.

Reason: It affects the kinetics of enzymes and through it the metabolic activity and other physiological functions of the organism.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

194. Match the following :

	Column-I		Column-II
(A)	Productivity	(I)	Rate of formation of new organic matter by consumers
(B)	Biosphere primary productivity	(II)	170 billion tons / year
(C)	GPP-R	(III)	$\text{gm}^{-2} \text{yr}^{-1}$
(D)	Secondary productivity	(IV)	NPP

	A	B	C	D
(1)	II	III	IV	I
(2)	III	II	I	IV
(3)	II	III	I	IV
(4)	III	II	IV	I

195. If we put a price tag on the ecosystem services ; then out of the total cost of various services, which service accounts for maximum cost ?

- (1) Nutrient cycling
- (2) Soil formation
- (3) Climate regulation
- (4) Habitat for wildlife

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196. Which is incorrect about ecological pyramids :

	Ecosystem	Biomass	Energy	Number
(1)	Tree			
(2)	Lake			
(3)	Grass Land			
(4)	Ocean			

197. **Statement-I** : India share of the global species diversity is an impressive 8.1 percent.

Statement-II : India has more than 50,000 genetically different strains of rice and 1000 varieties of mango.

- (1) Both statement-I and statement-II are correct.
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are incorrect

198. **Statement-I** : Amazon rain forest is estimated to produce 20% of the total oxygen in the earth's atmosphere.

Statement-II : Wildlife Safari Parks are an example of In-situ conservation.

- (1) Both statement-I and statement-II are correct.
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are incorrect

199. Chipko Movement was born in ?

- (1) 1974, Madhya pradesh
- (2) 1972, Uttarakhand
- (3) 1980, Rajasthan
- (4) 1974, Uttarakhand

200. In the tropical forests of different continents for frugivores (fruit-eating) birds and mammals value of Z found to be :-

- (1) 0.1 to 0.2
- (2) 0.6 – 1.2
- (3) 0.5 – 1.0
- (4) 1.15

		QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)	
		ENTHUSE ADV. COURSE : MED	
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
03 21-05-2023	SYLLABUS COVERED FROM 24-04-2023 TO 20-05-2023	P	NLM & FRICTION AND GRAVITATION
		C	CHEMICAL KINETICS
		B	HUMAN REPRODUCTION & REPRODUCTIVE HEALTH

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